

NGUYEN KHAC BAO

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Education

Posts and Telecommunications Institute of Technology (PTIT)

Bachelor of Information Technology

Ho Chi Minh City, Vietnam

2022 – Present

Relevant Coursework: Machine Learning, Deep Learning, Computer Vision

Skills & Technologies

Data & Annotation: YOLO/VOC labeling, 13-point keypoint annotation, multi-source dataset merging, class normalization, train/val/test splitting, dataset audit (distribution, malformed labels, duplicates, blur/dark flags), metadata & QA reports, version snapshots.

Programming: Python (NumPy, Pandas, Streamlit, PyYAML), C/C++ (basic).

CV Frameworks: PyTorch, Ultralytics YOLOv8/v11 (detection + pose), OpenCV, ByteTrack, EasyOCR, PaddleOCR.

Cloud: AWS (SageMaker, Lambda, S3), Google Cloud (Compute Engine, Cloud Storage, Vertex AI, Cloud Run).

Tools: Git/GitHub, Docker, Linux, Vast.ai, Jupyter, FastAPI, WebSocket.

CV Tasks: Object Detection, Pose Estimation, Video Analytics, Multi-object Tracking, Image Classification.

Languages: Vietnamese (native), English (technical reading proficiency).

Projects

FallGuard Labeling Tool — Computer Vision Dataset Assistant

2026

Python, Streamlit, OpenCV, YOLO-Pose, Pandas • [GitHub Repository](#)

- End-to-end dataset assistant: image/video ingestion, metadata tracking, auto pose-label generation, human review, QA reports, reviewed-only export.
- Streamlit review UI with bounding boxes, 13 keypoints, confidence scores, label correction, uncertain/exclude actions, manual keypoint editing for human-in-the-loop QA.
- Processed 5 videos into **656 reviewed frames**; versioned annotations, keypoint CSVs, event logs, QA reports, and validation snapshots.

Real-Time Face Mask Compliance Detection System

2026

Python, YOLOv8n, OpenCV, Vast.ai • [GitHub Repository](#)

- Converted PWMFD VOC→YOLO: **9,205 images**, **18,528 face instances** with class mapping, split validation, empty-label handling, SHA-1 reproducible naming.
- Generated dataset audit reports: per-split counts, per-class distribution, malformed-line detection to enforce data quality before training.
- Trained on Vast.ai: validation **mAP@50 0.970**, **mAP@50:95 0.749**; documented limited *incorrect_mask* sample size and per-class recall impact.
- Collected real-world test capture metadata (mask subtype, lighting, occlusion, blur, reflection) for structured edge-case analysis.
- Planned dataset extension: hand-occlusion subclasses (e.g., *correct_mask_hand_covered*, *incorrect_mask_hand_covered*, *no_mask_hand_covered*) to handle real-world camera scenarios where users partially obscure their face by hand.

Traffic Violation Detection & License Plate Recognition

2026

Python, YOLO, OpenCV, ByteTrack, OCR • [GitHub Repository](#)

- Fixed-camera video pipeline: vehicle detection/tracking (ByteTrack), traffic-light ROI state, stop-line crossing logic, violation events, plate OCR.
- Prepared CCPD subset of **8,598 images**: YOLO plate labels, OCR crops, train/val/test splits, negative samples, manifest metadata.
- Audit artefacts flagging missing files, invalid YOLO boxes, duplicates, dark/blur/small-plate samples, and contact sheets for human QA.

Activities

AI Research Group — University Club

Member

Ho Chi Minh City, Vietnam

2024 – Present

- Discussions and literature reviews on Machine Learning and Computer Vision (object detection, data augmentation).